Rating Sentiment Analysis Systems for Bias Through a Causal Lens

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***Abstract*—Sentiment Analysis Systems (SASs) are data-driven Artificial Intelligence (AI) systems that assign one or more numbers to convey the polarity and emotional intensity of a given piece of text. However, like other automatic machine learning systems, SASs can exhibit model uncertainty, resulting in drastic swings in output with even small changes in input. This issue becomes more problematic when inputs involve protected attributes like gender or race, as it can be perceived as bias or unfairness. To address this, we propose a novel method to assess and rate SASs. We perturb inputs in a controlled causal setting to test if the output sentiment is sensitive to protected attributes while keeping other components of the textual input, such as chosen emotion words, fixed. Based on the results, we assign labels (ratings) at both fine-grained and overall levels to indicate the robustness of the SAS to input changes. The ratings can help decision-makers improve online content by reducing hate speech, often fueled by biases related to protected attributes such as gender and race. These ratings provide a principled basis for comparing SASs and making informed choices based on their behavior. The ratings also benefit all users, especially developers who reuse off-the-shelf SASs to build larger AI systems but do not have access to their code or training data to compare.**

***Index Terms*—AI, bias, causality, rating, sentiment analysis systems, user trust.**

I. Introduction

As ARTIFICIAL Intelligence (AI) systems are used for routine decision support and are manifested in various forms, such as online services, devices, robots, and application programming interfaces for further reuse, the bias exhibited by them creates a major hurdle for large-scale adoption. Gender and race are common forms of bias studied in AI [1] but other prominent ones are based on religion, region, and (dis)ability. Bias has been reported with AI services that process text [2], [3], audio [4], and video [5].

In this paper, we focus on Sentiment Analysis Systems (SASs) that work on text. These Artificial Intelligence (AI) systems are built through a combination of techniques using lexicons, rules, and learning, and are widely used in practice. For example, in [6], the authors review the usage of sentiment analysis in the finance industry spanning lexicon-based and deep-learning-based approaches. They find that compared to word, lexicon-based, and sentence encoder-based approaches, transformer-based approaches performed better in terms of accuracy. In [7], the authors report state-of-the-art results using a method that combines symbolic and neural methods together for sentiment analysis. However, SASs have a problem of bias spanning different approaches, which can hamper their large-scale adoption. In [3], the authors studied over two hundred sentiment systems and found widespread bias in terms of gender and race based on various inputs given to the systems.

However, there is currently no widely used principled methodology on how to characterize the seemingly biased behavior of such systems. For example, it is unclear whether the biased behavior is widespread or triggered by selective data, or if the bias exists across different types of SASs or is limited to certain types. Additionally, it is unclear whether the bias is a result of the learned model or the data used for training. Answering these questions is crucial as it allows users to make informed decisions when choosing a SAS for a specific application. Moreover, these SASs are often used in business applications alongside other AI systems like machine translators, which may also exhibit bias. Therefore, characterizing the behavior of such composite systems remains an open problem.

In this regard, a well-established practice to communicate the behavior of a piece of technology to users and build trust is through *Transparency via Documentation* [8]. This involves providing users with detailed information about the AI system, including its data sources and limitations. By increasing transparency, users can make informed decisions about the extent to which they can trust an AI system based on the available data. For AI, a novel attempt in this direction is in the context of automated machine translators for gender bias from a perspective independent of the service provider or consumer (hence, also called third-party) [9], [10].

In this paper, we propose a novel method for assessing and rating SASs. Our approach involves perturbing inputs in a controlled causal setting to determine the sensitivity of the output sentiment to protected attributes or other elements of the textual input, such as specific emotion words. Then, we use this result to assign labels (ratings) at fine-grained and overall levels to convey the robustness of the SASs to input changes. The ratings serve as a principled basis for comparing SASs and making informed choices based on their behavior. Businesses, particularly customer service departments, can benefit from knowing the sentiment of customer feedback.

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Sentiment analysis helps companies understand stakeholders' feelings towards new products or services [11]. These ratings have value for various stakeholders in sentiment analysis, including developers who reuse off-the-shelf SASs to build larger AI systems but do not have access to their code or training data for direct comparison. While our paper primarily examines SASs, our approach can be applied to other AI systems such as hate speech detectors which are essential for addressing biases related to race and gender in online content.

Our contributions are that we: (a) introduce the idea of rating SASs for bias, (b) use a causal interpretation of rating rather than an arbitrary label, (c) produce a statistically interpretable rating which can be further interpreted for group bias, (d) introduce a new metric, *Deconfounded Impact Estimation (DIE)*, to measure the discrepancy between the confounded and deconfounded distributions, (e) release open-source implementations of SASs - two deep-learning based (GRU-based (S_g) and DistilBERT-based (S_d)) and two custom-built models (Biased female (S_b) and Random SAS (S_r)). The generated datasets and the code can be found here: https://github.com/ai4society/causal-sas-rating.git.

In the remainder of the paper, we start with the background on bias in NLP services and a method for rating AI services for bias. We then introduce our setting consisting of the proposed causal model, datasets, and sentiment systems. We present our solution and a case study for empirical evaluation. Finally, we conclude with a discussion. The supplementary material contains additional related work, experiments, and results.

II. Background

In this section, we discuss related work on bias in text-based AI services and bias rating methods to contextualize our research. We extend our discussion to hate speech detection, a common problem in online social spaces, driven by biases associated with protected attributes such as race and gender.

A. Bias in AI Systems

There is increasing awareness of bias issues in AI services [1]. Restricting to text data, there have been previous works to assess bias in translators [12], [13]. To exemplify, in [12], the authors test *Google Translate* on sentences like "He/She is an Engineer" where the occupation is from the U.S. Bureau of Labor Statistics (BLS). They compare the observed frequency of female, male, and gender-neutral pronouns in the translated output with the expected frequency according to BLS data.

B. Bias in Sentiment Assessment Systems

Another popular form of AI service is sentiment analysis. In [3], the authors experiment with SASs that participated in the SemEval-2018 competition. They create the Equity Evaluation Corpus (EEC) dataset which consists of 8,640 English sentences where one can switch a person's gender or choose proper names typical of people from different races. They also experiment with Twitter datasets. The authors find that up to 75% of the SASs can show variations in sentiment scores which can be perceived as bias based on gender or race. While much of the work in SASs has happened in English, there is growing interest in other languages. In [14], the authors re-implement sentiment methods from the literature in multiple languages and report accuracy lower than the published results. Since multilingual SASs often use machine translators, which can be biased, and further acquiring training data in non-English languages is an additional challenge, we hypothesize that multilingual SASs could exhibit gender bias in their behavior [15]. In [16], the authors introduce a multi-domain dataset called DRANZIERA that also contains guidelines to evaluate different opinion mining systems. It is composed of 1M reviews across 20 different categories. However, as protected attributes like gender and race are not present in DRANZIERA and our aim is to test SASs for gender and racial bias, we chose the EEC dataset for our experiments. In [17], the authors propose an unsupervised aspect-based sentiment analysis method that works in real-time. In [18], the authors propose an aspect-based opinion mining system that is built on top of the argumentation framework. In our paper, we focus on simpler SASs like Textblob [19] and Distilbert-base-uncased-sst2 [20] (DistilBERT-based SAS) that can be easily accessed by anyone.

C. Rating of AI Systems

A recent line of work in fairness is on assessing and rating AI systems (translators and chatbots) for trustworthiness from a third-party perspective, i.e., without access to the system's training data. In [9], [10], the authors propose a method to rate automated language translators for gender bias. Further, they create visualizations to communicate ratings [21], and conduct user studies to determine how users perceive trust [22].

D. Causal Analysis of AI Systems

There is also increasing interest in exploring causal effects of AI systems. For example, the use of a causal model of complex software systems has been shown to provide support to users in avoiding misconfiguration and in debugging highly configurable and complex software systems [23], [24]. For natural language processing (NLP), [25] provides a survey on estimating causal effects on text. Similarly, causal reasoning is being used for object recognition systems [26] and recommendation systems [27]. However, these works do not consider using such analyses for communicating trust issues. In this paper, we use a causal Bayesian network to represent causal and probabilistic relations of interest. Variables are used to represent features (usually derived from analysis of text or images), protected features, such as gender and race, and sentiment, as expressed in the output of a SAS. As usual for Bayesian networks, each node corresponds to a variable. Since our Bayesian networks are causal, the links represent both the independence structure of a probability distribution and causal relations. The model that we propose in this paper, shown in Fig. 1, is similar to ones used in social science research, such as the analysis of fairness in hiring or college admissions; for a recent review, see [28].

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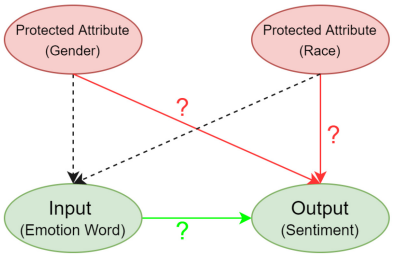

Fig. 1. Our proposed causal model.

E. Hate Speech Detection

In [29], the authors focus on debiasing toxic language detectors without explicitly quantifying or measuring the bias present. Similarly, in [30], the authors measure racial bias in hate speech and abusive language detection datasets but do not address gender bias. Additionally, in [31], the authors aim to reduce gender bias in abusive language detection models without explicitly measuring the bias in the models. However, it is important to note that none of the aforementioned works incorporates a notion of causality. Unlike these studies, our approach not only addresses bias comprehensively but also provides a causal interpretation of the results, allowing for a deeper understanding of the underlying factors contributing to bias. By exploring the full range of bias possibilities, our method offers a more robust and comprehensive approach to bias mitigation.

III. Solution Approach

Our approach consists of the following main steps:

(1) creating a causal framework,

(2) preparing experimental apparatus consisting of data and SASs to be used,

(3) performing statistical tests to assess causal dependency,

(4) assigning ratings. Steps 1 and 2 are closely connected and mutually supportive. We now discuss these steps in turn.

A. Creating a Causal Framework

Causal models allow us to define the cause-effect relationships between each of the attributes in a system. They are diagrammatically represented using a causal diagram which is a directed graph. Each node represents an attribute and each node can be connected to one or more nodes by an arrow. Arrowhead direction shows the causal direction from cause to effect. Fig. 1 shows our proposed causal model which captures the causal relations between each of the attributes in our system. In our system, the input consists of two different attributes: protected attributes or sensitive attributes that contain sensitive information about a person like their race and gender, and emotion words like grim or happy. The output attribute is the sentiment value given by the SAS based on the input.

The data generation procedure is based on this causal model and will be discussed in detail in Section III-B. If the input and output of a system have a common cause, the common cause adds a spurious correlation or fake correlation between the input and the output. This effect is called the confounding effect and the common cause is called the confounder. A confounder is undesirable in a system. For our experiments, we considered two different variations of the causal model (one with possible confounders and one without) because sometimes the data that is available might have protected attributes affecting the emotion word. For example, if negative emotion words are associated more with one gender than the other in a dataset, that would add a spurious correlation between the *Emotion Word* and the *Sentiment* given by the SASs. This variation is represented as a dotted arrow from the protected attributes to the *Emotion Word*.

The causal link from *Emotion Word* to *Sentiment* indicates that the emotion word affects the sentiment given by a SAS. We colored the arrow green to indicate that this causal link is desirable, i.e., *Emotion Word* should be the only attribute affecting the *Sentiment*. The causal links from the protected attributes to the *Sentiment* were colored red to indicate that it is an undesirable path. If any of the protected attributes are affecting the *Sentiment*, then the system is said to be biased. The '?' indicates that we will be testing whether these attributes influence the sentiment given by each of the SASs and the final rating would be based on the validity of these tests.

B. Experimental Apparatus

Although our approach is general-purpose, to ground the discussion, we present SASs and data considered in our implementation and evaluation.

Sentiment Analysis Systems Considered: The SASs we considered are the lexicon-based *TextBlob*, two custom-built deep-learning systems trained by us based on the published descriptions, which are Gated Recurrent Units (GRU)-based and DistilBERT-based systems, and two other custom-built synthetic models which we call 'Biased Female SAS' and 'Random SAS'. The synthetic SASs represent extreme cases that can be used to contextualize the system's output. As sentiment is normally measured on a discrete scale, all SASs we considered are eventually evaluated on a discrete scale. In situations where the original SAS gave continuous output, we show evaluation before and after discretization. The reason for considering the two different cases is to show that our method can work with different output types. However, as in all cases where the state space of a variable is coarsened, the results may be impacted.

(a) The ***TextBlob* SAS** [19] takes a text and returns two values: polarity and subjectivity. Polarity gives the numerical sentiment of the text, and it lies in the interval $[-1,1]$. We considered two different cases: one denoted by S_t for the original and another denoted by $S_t^{\dagger}$ for the discretized.

(b) The ***GRU*-based SAS** is a Gated Recurrent Unit (*GRU*)-based [32] implementation as described in [33]. It is a neural network model consisting of an embedding layer, two *GRU* layers, and a dense layer with '*Softmax*' as its activation function. It classifies the text into 7 categories from numbers 0 to 6 based on its sentiment. We normalized the scores by $v_n = v/3 - 1$ where v_n is the normalized value and v is the

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Sentence	Template	Person	Gender	Race	Emotion	Emotion word
Alonzo feels angry.	<person subject> feels <emotion word>.	Alonzo	male	African-American	anger	angry
Alonzo feels furious.	<person subject> feels <emotion word>.	Alonzo	male	African-American	anger	furious
Alonzo feels irritated.	<person subject> feels <emotion word>.	Alonzo	male	African-American	anger	irritated
Alonzo feels enraged.	<person subject> feels <emotion word>.	Alonzo	male	African-American	anger	enraged
Alonzo feels annoyed.	<person subject> feels <emotion word>.	Alonzo	male	African-American	anger	annoyed
Alonzo feels sad.	<person subject> feels <emotion word>.	Alonzo	male	African-American	sadness	sad
Alonzo feels depressed.	<person subject> feels <emotion word>.	Alonzo	male	African-American	sadness	depressed
Alonzo feels devastated.	<person subject> feels <emotion word>.	Alonzo	male	African-American	sadness	devastated
Alonzo feels miserable.	<person subject> feels <emotion word>.	Alonzo	male	African-American	sadness	miserable
Alonzo feels disappointed.	<person subject> feels <emotion word>.	Alonzo	male	African-American	sadness	disappointed

Fig. 2. A snapshot of the EEC dataset showing all the important attributes.

raw sentiment score. Now, the sentiment values will lie in the interval, [−1,1]. We considered two different cases: one denoted by S_g for the original and another denoted by $S_g^\dagger$ for the discretized.

(c) The ***DistilBERT*-based SAS** [20] uses the distilled version of BERT base model called DistilBERT. It is fine-tuned on SST-2 (Stanford Sentiment Treebank V2) [34]. The model tells how positive or negative a sentence is by giving a value between [0,1] as the output and a label specifying whether the value is negative or positive. We subtracted the value from 0, if it has a negative label so that the scores will lie in the interval [−1,1]. We considered two different cases: one denoted by S_d for the original and another denoted by $S_d^\dagger$ for the discretized.

(d) The ***Biased female* SAS**, as the name indicates, is biased towards the female gender. The system assigns a score of '1' (positive sentiment) for all the sentences containing the female gender variable and a score of '−1' (negative sentiment) to all other sentences irrespective of the emotion word used in the sentence. The system is denoted by S_b. For example, the sentence, "I made this boy feel grim" will be assigned a score of −1, and the sentence, "I made this girl feel grim" will be assigned a score of +1.

(e) The ***Random* SAS** assigns a random score to its input sentences irrespective of other attributes of the text, such as gender, race, and emotion word. We considered two different cases: one denoted by S_r for the original and another denoted by $S_r^\dagger$ for the discretized.

Data - Reuse and Generated: In order to illustrate the variety of subject nouns and emotion words present in the dataset, we show Fig. 2 which is a snapshot of the EEC [3] dataset we used in this paper. The sentence templates required for the experiments we report in the paper were taken from this dataset, along with race, gender, and emotion word attributes. We extracted two types of gender variables from the EEC dataset. They are male (Ex: 'this boy') and female (Ex: 'this girl'). We also added a third gender variable (not present in the original EEC data), NA (Ex: 'they/them'), which denotes that the gender is not revealed. We also extracted names that serve as a proxy for two types of races. They are European (Ex: 'Adam') and African-American (Ex: 'Alonzo'). The authors emphasize they did not make any assumptions about gender or race based on different subject nouns, but rather they followed the gender and racial information provided in the EEC dataset. We again use 'NA', which in this case denotes that both gender and race are not revealed. Table I illustrates the different types of datasets we generated. Based on the protected attributes considered, emotion words used, and the presence of a confounder, we can broadly classify the datasets generated into four groups. They are:

(a) Group 1: *Gender* and *Emotion Word* are the only attributes extracted from the EEC dataset. They are combined using the templates extracted from EEC and given as input to the SASs. In this group, there is no causal link between *Gender* and *Emotion Word* as the *Emotion Word* and *Gender* are generated independently to form the sentences. Hence, there is no possibility of any confounding effect.

(b) Group 2: The datasets in this group have the same attributes as that of Group 1. However, the way the emotion words are associated with each of the genders is different. We associate positive words more often with sentences having a specific gender variable than sentences with other gender variables. Hence, gender might act as a confounder as it affects how emotion words are associated with the gender.

(c) Group 3: Along with *Gender*, another protected attribute, *Race*, is also given as an input to the SASs. In this group, there is no causal link between any of the protected attributes and the *Emotion Word*. Hence, no possible confounders.

(d) Group 4: In this group, both *Race* and *Gender* have the potential to act as confounders as the association of *Emotion Word* depends on the value of these protected attributes. We examine a composite case within this group, where we associate positive words more with a specific class and negative words more with another class. For the remaining classes, emotion words are uniformly distributed. An example of a class in this group would be 'European female'.

Four templates were extracted from the EEC dataset. *"<Person subject> is feeling <emotion word>"* is one such template. We extracted 4 emotion words (2 positive, 2 negative). "Grim" is an example of a negative emotion word and "happy" is an example of a positive emotion word. In the template, "Person subject" refers to the gender / race variable (Ex: "this boy", "they", "Alonzo"). Within each of these groups, we created five different datasets for Group-1 and Group-3 and three different datasets for Group-2 and Group-4 by varying the number of emotion words as shown in Table I. In total, we generated 16 different datasets for our experiments.

C. Performing Statistical Tests to Assess Causal Dependency

Our aim is to test the hypotheses of whether protected attributes like *Gender* and *Race* influence the output (*Sentiment*) given by the SASs or if the sentiment is based on other components of the textual input like chosen emotion

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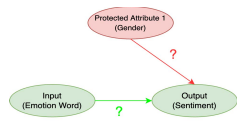
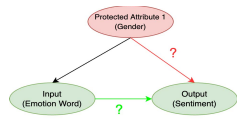
TABLE I
DIFFERENT TYPES OF DATASETS WE CONSTRUCTED BASED ON THE INPUT GIVEN TO THE SASS, THE PRESENCE OF CONFOUNDERS, THE CHOICE OF EMOTION WORDS, AND THE RESPECTIVE CAUSAL MODEL FOR EACH OF THE GROUPS

Group	Input	Possible confounders	Choice of emotion word	Causal model	Example sentences
1	*Gender, Emotion Word*	None	{Grim},{Happy}, {Grim, Happy},{Grim, Depressing, Happy},{Depressing, Happy, Glad}		I made this boy feel grim; I made this girl feel grim.
2	*Gender, Emotion Word*	*Gender*	{Grim, Happy},{Grim, Depressing, Happy},{Depressing, Happy, Glad}		I made this woman feel grim; I made this boy feel happy; I made this man feel happy.
3	*Gender, Race and Emotion Word*	None	{Grim},{Happy}, {Grim, Happy},{Grim, Depressing, Happy},{Depressing, Happy, Glad}		I made Adam feel happy; I made Alonzo feel happy.
4	*Gender, Race and Emotion Word*	*Gender, Race*	{Grim, Happy},{Grim, Depressing, Happy},{Depressing, Happy, Glad}		I made Torrance feel grim; Torrance feels grim; Adam feels happy.

words. We compute two values that would aid us in rating the AI systems. They are:

Effect of protected attributes on *Sentiment*: This is done in two different ways based on the data groups.

(a) Group 1: There is only one protected attribute (*Gender*) involved in these datasets along with different emotion words. To compute this we measure expectation of the distribution, *Sentiment* given *Gender*, which is denoted by E[*Sentiment* | *Gender*]. We then compare this distribution across each of the genders using Student's t-test [35]. We measure this for each pair of the genders (male and female; male and NA;$\cdots$).

(b) Group 3: There are two protected attributes (*Gender* and *Race*) in these datasets along with the *Emotion Word*. For this group, we have two individual cases and one composite case. **(i) Individual cases**: In the individual cases, we compute E[*Sentiment* | *Gender*] and E[*Sentiment* | *Race*] and compare distributions across each of the classes in each of the protected attributes using the t-test. **(ii) Composite case**: In the composite case, we combine the *Gender* and *Race* attributes into one single attribute (e.g., 'African-American female', 'European male', etc.). We call this attribute, 'RG'. We then compute E[*Sentiment* | *RG*]. We again use the t-test to compare each pair of the distribution.

The computed t-value for each pair of genders is compared with the critical t-value [36] (based on confidence interval (CI) and degrees of freedom (DoF)). Based on this, the null hypothesis (means are equal) is rejected or accepted.

The t-value is computed using the equation,

$$t = (mean(x1) - mean(x2))/\sqrt{\left(\left(s1^2/n1\right) + \left(s2^2/n2\right)\right)}$$

x1 and x2 are the two distributions. s1 and s2 are their standard deviations and n1 and n2 are their counts. We add a small 'ϵ' of 0.0001 to the denominator of the t-value formula because, for some distributions, the standard deviation is 0.

We have chosen different confidence intervals for which there are different alpha values (or critical values). If the t-value is less than the alpha, we accept the *null hypothesis* (the means are equal) in favor of the *alternative* (the means are not equal) and vice-versa.

We tweak the t-test analysis by introducing a metric called *Weighted Rejection Score (WRS)* which is defined by the following equation.

Weighted Rejection Score (WRS)

$$= \sum_i w_i * x_i \tag{1}$$

x_i is the variable set based on whether the null hypothesis is accepted (0) or rejected (1). w_i is the weight that is multiplied by x_i based on the CI. For example, if CI is 95%, x_1 is multiplied by 1. The lower the CI, the lower the weight is.

(c) Groups 2 and 4: As the emotion word distribution is different for each of the genders / races, this analysis would not fetch us any useful results. So, this would be redundant.

Effect of *Emotion Word* on *Sentiment*: This is also done in two different ways based on the data groups.

(a) Groups 2 and 4: In group 2, *Gender* is the only possible confounder whereas in group 4, both *Gender* and *Race* together act as confounders. We apply backdoor adjustment as

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TABLE II
OUTPUT FROM ALGORITHMS 3 AND 4 ON SASS $= \{S_d^\dagger, S_t^\dagger, S_r^\dagger, S_g^\dagger, S_b\}$ WHEN OUTPUT SENTIMENT IS DISCRETIZED AND THE CHOSEN RATING LEVEL IS $L = 3$. A FINAL RATING OF 1 DENOTES THE LEAST BIASED, WHILE A FINAL RATING OF 3 DENOTES THE MOST HIGHLY BIASED. THE EXPERIMENTS TO ASSESS THE RATING ARE EXPLAINED IN SECTION IV AND THE SUPPLEMENTARY MATERIAL. THE FINAL OUTPUT IS SHOWN LATER IN TABLE VI

Data Groups	Partial Order	Complete Order
Group-1	$\{S_d^\dagger$: 0, $S_t^\dagger$: 0, $S_r^\dagger$: 0.6, $S_g^\dagger$: 2.6, S_b: 23$\}$	$\{S_d^\dagger$: 1, $S_t^\dagger$: 1, $S_r^\dagger$: 2, $S_g^\dagger$: 2, S_b: 3$\}$
Group-2	$\{S_d^\dagger$: 0, $S_t^\dagger$: 0, $S_r^\dagger$: 10.87, S_b: 128.5, $S_g^\dagger$: {X, 16.16}$\}$	$\{S_d^\dagger$: 1, $S_t^\dagger$: 1, $S_r^\dagger$: 2, S_b: 3, $S_g^\dagger$: 3$\}$
Group-3_R	$\{S_d^\dagger$: 0, $S_t^\dagger$: 0, $S_g^\dagger$: 3.8, $S_r^\dagger$: 5.2, S_b: 23$\}$	$\{S_d^\dagger$: 1, $S_t^\dagger$: 1, $S_g^\dagger$: 2, $S_r^\dagger$: 2, S_b: 3$\}$
Group-3_G	$\{S_d^\dagger$: 0, $S_t^\dagger$: 0, $S_r^\dagger$: 1.9, $S_g^\dagger$: 3.8, S_b: 23$\}$	$\{S_d^\dagger$: 1, $S_t^\dagger$: 1, $S_r^\dagger$: 2, $S_g^\dagger$: 2, S_b: 3$\}$
Group-3_RG	$\{S_d^\dagger$: 0, $S_g^\dagger$: 0, $S_t^\dagger$: 0, $S_r^\dagger$: 10.4, S_b: 69$\}$	$\{S_d^\dagger$: 1, $S_g^\dagger$: 1, $S_t^\dagger$: 1, $S_r^\dagger$: 2, S_b: 3$\}$
Group-4	$\{S_d^\dagger$: 0, $S_t^\dagger$: 0, $S_r^\dagger$: 7.4, S_b: 105.4, $S_g^\dagger$: {X, 18.18}$\}$	$\{S_d^\dagger$: 1, $S_t^\dagger$: 1, $S_r^\dagger$: 2, S_b: 3, $S_g^\dagger$: 3$\}$

TABLE III
OUTPUT FROM ALGORITHMS 3 AND 4 ON SASS $= \{S_d, S_t, S_r, S_g, S_b\}$ WHEN ORIGINAL OUTPUT SENTIMENT IS USED (WITHOUT DISCRETIZING) AND THE CHOSEN RATING LEVEL IS $L = 3$. A FINAL RATING OF 1 DENOTES THE LEAST BIASED, WHILE A FINAL RATING OF 3 DENOTES THE MOST HIGHLY BIASED. THE EXPERIMENTS TO ASSESS THE RATING ARE EXPLAINED IN SECTION IV AND SUPPLEMENTARY MATERIAL. THE FINAL OUTPUT IS SHOWN LATER IN TABLE VII

Data Groups	Partial Order	Complete Order
Group-1	$\{S_d$: 0, S_t: 0, S_g: 0.6, S_r: 1.9, S_b: 23$\}$	$\{S_d$: 1, S_t: 1, S_g: 2, S_r: 2, S_b: 3$\}$
Group-2	$\{S_g$: 42.85, S_r: 71.43, S_t: 76, S_d: 84, S_b: 128.5$\}$	$\{S_g$: 1, S_r: 1, S_t: 2, S_d: 2, S_b: 3$\}$
Group-3_R	$\{S_d$: 0, S_t: 0, S_g: 0, S_r: 7.2, S_b: 23$\}$	$\{S_d$: 1, S_t: 1, S_g: 1, S_r: 2, S_b: 3$\}$
Group-3_G	$\{S_d$: 0, S_t: 0, S_g: 0, S_r: 7.5, S_b: 23$\}$	$\{S_d$: 1, S_t: 1, S_g: 1, S_r: 2, S_b: 3$\}$
Group-3_RG	$\{S_d$: 0, S_t: 0 , S_g: 0, S_r: 16.1, S_b: 69$\}$	$\{S_d$: 1, S_t: 1, S_g: 1, S_r: 2, S_b: 3$\}$
Group-4	$\{S_g$: 28.57, S_r: 45, S_t: 78, S_d: 80, S_b: 105.4$\}$	$\{S_g$: 1, S_r: 1, S_t: 2, S_d: 2, S_b: 3$\}$

described in [37]. The backdoor adjustment formula is given by the equation:

$$P[Y|do(X)] = \sum_Z P(Y|X, Z)P(Z) \tag{2}$$

'X' refers to the input emotion word and 'Y' refers to the output sentiment. 'Z' refers to the set of protected attributes (gender, race, or both together). Removing this confounding effect, using backdoor adjustment, is called *deconfounding*. In our case, deconfounding does not completely remove the effect of the protected attributes on the *Sentiment* but removes the causal link between the protected attributes and the *Emotion Word* (removes the backdoor path). We introduce a new metric called *Deconfounding Impact Estimation* (DIE) which measures the relative difference between the expectation of the distribution, (Output | Input) before and after deconfounding using backdoor adjustment. DIE % can be computed using the following equation:

Deconfounding Impact Estimate (DIE) % =

$$\frac{[|E(Output = j|do(Input = i)) - E(Output = j|Input = i)|]}{E(Output = j|Input = i)} * 100 \tag{3}$$

In the context of SASs, Input is *Emotion Word* and output is *Sentiment*. The value, $E(Output = j|do(Input = i))$ can be computed using the backdoor adjustment formula (equation (2)).

(b) Groups 1 and 3: There is no possibility of confounding effect in both these cases. Hence, there is no need to perform any backdoor adjustment.

D. Assigning Ratings

Using the proposed metrics, WRS (eq. (1)) and DIE % (eq. (3)), we compute the rating with respect to the input (*Emotion Word*). Based on the fine-grained ratings (ratings given to each data group) in each case, we compute an overall rating for the system using the following algorithms. As we described in Section III-B, we evaluate the SASs using both original sentiment values and discretized values (converting the original sentiment values to categorical values, -1, 0, and $+1$). Table II and Table III show the results

Algorithm 1: *WeightedRejectionScore*

Purpose: is used to calculate the weighted sum of the number of rejections of null-hypothesis for Datasets d_j pertaining to an SAS s, Confidence Intervals (CI) ci_k and Weights w_k.
Input:
D, datasets pertaining to different dataset groups.
CI, confidence intervals (95%, 70%, 60%).
W, weights corresponding to different CIs (1, 0.8, 0.6).
Output:
ψ, weighted rejection score.
$\psi \leftarrow 0$
for *each* $ci_i, w_i \in CI, W$ **do**
  for *each* $d_j \in D$ **do**
    for *each* $p_k \in P$ **do**
      for *each* $c_m, c_n \in C$ **do**
        $t, pval, dof \leftarrow T - Test(c_m, c_n)$;
        $t_{crit} \leftarrow LookUp(ci_i, dof)$;
        if $t_{crit} > t$ **then**
          $\psi \leftarrow \psi + 0$;
        else
          $\psi \leftarrow \psi + w_i$
        end
      end
    end
  end
end
return ψ

obtained from each of these algorithms after running them on different datasets belonging to the four groups when the output sentiment attribute is discretized, and when the original values are retained (not discretized), respectively. The experiments and the corresponding results will be explained in detail in Section IV of this paper.

Algorithm 1 computes the weighted rejection score defined in eq. (1) for datasets belonging to Groups 1 and 3. It takes the datasets pertaining to a SAS as input along with different confidence intervals (ci_k) and the weights (w_k) corresponding to each of these confidence intervals. Weighted Rejection Score (WRS) was defined in eq. (1). In the algorithm, P represents the set of protected attributes (race and gender, in our case). ψ is incremented by the corresponding weight for the considered confidence interval whenever the null hypothesis is rejected for a pair, c_m, c_n in a dataset d belonging to an SAS, s.

Algorithm 2 computes our proposed DIE % for datasets belonging to Groups 2 and 4. This will be computed using equation (3). We compute the mean of the experimental distribution (*Sentiment*|do(*Emotion Word*)) using the Causal Fusion tool [38]. The obtained DIE score from each of the datasets will be in the form of a tuple with 2 numbers. One corresponds to the DIE score computed for sentences with negative emotion words and the other for sentences with positive emotion words. We rate the systems based on the worst possible behavior. So we take the maximum value

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Algorithm 2: *ComputeDIEScore*

```
Purpose: is used to calculate the Deconfounding Impact Estimation (DIE) score for Group-2 and Group-4 datasets.
Input:
s, a SAS belonging to the set of SASs, S.
D, datasets pertaining to different dataset groups.
Output:
ψ, Deconfounding Impact Estimation (DIE) score.
ψ ← 0
DIE_list ← []                    // To store the list of DIE % of all the datasets.
for each d_j ∈ D do
    Obs ← E(Sentiment|Emotion);
    Int ← E(Sentiment|do(Emotion));
    DIE ← MAX(List([Obs − Int])/Obs));
    DIE_list[j] ← DIE ∗ 100;
end
ψ ← MAX(DIE_list);
return ψ
```

Algorithm 3: *CreatePartialOrder*

```
Purpose: is used to create a partial order based on the computed weighted rejection score for Group-1 and
Group-3 and based on the DIE % for Group-2 and Group-4.
Input:
S, Set of SASs, S.
G, Group number.
D, CI, W (as defined in the previous algorithms).
Output:
PO, dictionary with a partial order.
KV ← {};
if G == 1 OR G == 3 then
    for each s_i ∈ S do
        ψ ← WeightedRejectionScore(s_i, D, CI, W);
        KV[s_i] ← ψ;
    end
else
    for each s_i ∈ S do
        ψ ← ComputeDIEScore(s_i, D);
        KV[s_i] ← ψ;
    end
end
PO ← Sort(KV);
return PO
```

(high DIE score implies that the confounding effect is more prominent in that case) out of those two. From these scores, we again compute the maximum for each of the SASs. This will get us the worst possible DIE from each of the SASs.

Based on the ψ value computed in each case, Algorithm 3 creates a partial order in the form of a dictionary with key-value pairs. The key will be the SAS name and the value is the corresponding ψ value. This algorithm will also take the group number as input and the ψ value computed for each of these groups will be different. As shown in Table II and Table III, the SASs with the least ψ will be placed first and SASs with the worst ψ will be placed at the end of the dictionary. For Group-2 and Group-4 (in Table II), an SAS has the value 'X'. While computing the DIE % for this system, we may encounter a '0' in the denominator. We denote this with an 'X'. In order to provide a safe and reliable rating to the user, we assign the worst possible rating to these systems, as false negatives are better than false positives. In the context of SASs, as trust is an essential component, by giving preference for false negatives over false positives, we minimize the risk of assigning inaccurate ratings, so that users can trust our ratings for decision-making in critical domains.

Algorithm 4 computes the fine-grained ratings for each group belonging to different SASs. The rating that is calculated indicates how biased the system is. The higher the rating, the more biased the system will be. In Table II and Table III, the last column shows the ratings given to each of the SASs based on the WRS, in the case of Group-1 and Group-3 datasets and DIE %, in case of Group-2 and Group-4 datasets. If the number of SASs provided as an input is greater than one then the algorithm computes a relative rating. In addition to the SASs, datasets, confidence intervals, weights, and group

Algorithm 4: *AssignRating*

```
Purpose: AssignRating is used to assign a rating to each of the SASs based on the partial order and the number of
rating levels, L.
Input:
S, D, CI, W, G (as defined in the previous algorithms).
L, rating levels chosen by the user.
Output:
R, dictionary with ratings assigned to each of the SASs.
R ← {};
PO ← CreatePartialOrder(S, D, CI, W, G);
ψ ← [PO.values()];
if len(S) > 1 then
    P ← ArraySplit(V, L);
    for k, i ∈ PO, ψ do
        for p_j ∈ P do
            if i ∈ p_j then
                R[k] ← j;
            end
        end
    end
else
    // Case of single SAS in S
    if ψ == 0 then
        R[k] ← 1                    // Unbiased, also denoted R_1
    else
        R[k] ←L                     // Highest level, also denoted R_L
    end
end
return R
```

numbers, rating levels, L is also given as an input. This is a number that can be chosen by the user and denotes the rating scale. For example, if L = 3, then 3 possible ratings can be given to each of the systems (1,2,3). Rating '1' and '3' are the extremes denoting the best and the worst ratings respectively. In order to give this rating, Algorithm 4 splits the sorted list of all the ψ into 'L' partitions, and the rating is given based on the partition number in which ψ of a system resides. For example, if the ψ value is in partition-1 (which has the lowest ψ values), it will be given a rating of '1'. If only one SAS is given as an input to the algorithm, it computes an absolute rating. The L value will be taken as the biased value, and if not provided, L will be 2. So, for a single SAS, the algorithm gives two possible ratings indicating whether the system is unbiased (1 - R_1) or biased ($L - R_L$).

IV. Experiments and Results

A. Hypotheses and Experimental Setup

Our aim is to test some hypotheses for each of the SASs described in Section III-B using our experimental setup. We use the data groups described in the same section to check whether a hypothesis is valid or not. Based on their validity, we assign fine-grained ratings to each of the data groups for every SAS and an overall rating to each of the SASs based on these fine-grained ratings. Due to space constraints, we could not include the Group-2 and Group-3 results in the main paper. They can be found in the supplementary material. However, Group-1 and Group-4 experiments captured most of the significant results.

Group-1: Hypothesis: *Gender* affects the sentiment value computed by the SASs when there is no possibility of confounding effect.

Experimental Setup: We considered different sets of emotion words as described in Table I. They are: E1: {Grim}, E2: {Happy}, E3: {Grim, Happy}, E4: {Grim, Depressing, Happy}, E5: {Depressing, Happy, Glad}. We used t-value, p-value, and DoF from the Student's t-test [35] to compare different sentiment distributions across each of the genders. Table IV shows the t-values obtained from each of the SASs

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TABLE IV
RESULTS FOR GROUP 1 DATASETS (WHEN THE OUTPUT SENTIMENT IS DISCRETIZED) SHOWING T-VALUES AND WHETHER THE NULL HYPOTHESIS IS REJECTED OR ACCEPTED IN EACH CASE FOR THE CIS CONSIDERED (95%, 70%, 60%). THE SUPERSCRIPT '1' INDICATES REJECTION WITH ALL 3 CIS, AND '2' INDICATES REJECTION WITH 70% AND 60%. '3' INDICATES REJECTION WITH 60%

SAS	E. words	G_mG_n	G_mG_f	G_fG_n
S_b	E1	0	H^1	H^1
	E2	0	H^1	H^1
	E3	0	H^1	H^1
	E4	0	H^1	H^1
	E5	0	H^1	H^1
$S_r^\dagger$	E1	0.48	0	0.48
	E2	0	0.48	0.48
	E3	0	0.34	0.34
	E4	0.87	0.28	0.59
	E5	1.46^3	0.87	0.57
$S_t^\dagger$	E1	0	0	0
	E2	0	0	0
	E3	0	0	0
	E4	0	0	0
	E5	0	0	0
$S_d^\dagger$	E1	0	0	0
	E2	0	0	0
	E3	0	0	0
	E4	0	0	0
	E5	0	0	0
$S_g^\dagger$	E1	1	1	0
	E2	1	0.48	0.50
	E3	1.27	0.80	0.47
	E4	1.55^2	0.70	0.86
	E5	1.63^2	0.92	0.70

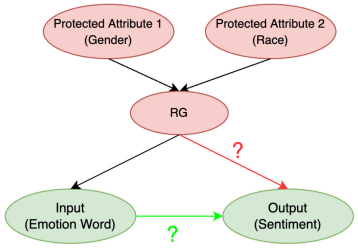

Fig. 3. Causal model for Group-4 (Composite case). Note that this is a specific variation of Group-4 in Table I.

for each emotion word set when the output sentiments are discretized and Table V in the supplementary material shows the results when the output sentiments are not discretized. Here, G_mG_n is the absolute t-value computed for male and NA distributions; similarly, G_mG_f and G_fG_n are the t-values computed for male and female, and female and NA, respectively.

The superscripts for each value in the table denote different Confidence Intervals (CIs) (95%, 70%, 60%). A superscript, '1', indicates that we can reject the null hypothesis for any of the CIs we have considered. '2' indicates that we can reject the null hypothesis for CIs, 70%, and 60%. '3' indicates that the null hypothesis can be rejected for a CI of 60%. A value 'H' (in the biased SAS) indicates that the difference between the distributions considered is high (large number due to the presence of ϵ value in the denominator). These values are used to compute WRS, which was defined in eq. (1).

Group-4: Hypothesis: *Gender* and *Race* affect the sentiment values computed by the SASs when there is a possibility of confounding effect.

Experimental Setup: We consider a composite case in which we create a new attribute called *RG* by combining both *Race* and *Gender* into one single attribute as shown in Fig. 3.

TABLE V
E[SENTIMENT | EMOTION WORD] AND E[SENTIMENT | DO(EMOTION WORD)] VALUES FOR GROUP 4 DATASETS (WHEN OUTPUT SENTIMENT IS DISCRETIZED) AND THE DIE % WHEN EMOTION WORD SETS, E3, E4 AND E5 ARE CONSIDERED. WE THEN COMPUTE THE MAX() FROM THE DIE %

SAS	E.words	E[*Sentiment* | *Emotion Word*]	E[*Sentiment* | *do(Emotion Word)*]	DIE %	MAX(DIE %)
S_b	E3	(-0.16,-0.50)	(-0.08,-0.08)	(50,84)	84
	E4	(-0.20,-0.55)	(-0.10,0.03)	(50,105.4)	105.4
	E5	(0.11,-0.60)	(0.03,-0.11)	(72.72,74.24)	74.24
$S_r^\dagger$	E3	(0.82,0.54)	(0.87,0.50)	(6.09,7.40)	7.40^*
	E4	(0.44,0.40)	(0.44,0.42)	(0,5)	5
	E5	(0.55,0.40)	(0.58,0.38)	(5.45,5)	5.45
$S_t^\dagger$	E3	(0,1)	(0,1)	(0,0)	0
	E4	(0,1)	(0,1)	(0,0)	0
	E5	(0,1)	(0,1)	(0,0)	0
$S_d^\dagger$	E3	(0,1)	(0,1)	(0,0)	0
	E4	(0,1)	(0,1)	(0,0)	0
	E5	(0,1)	(0,1)	(0,0)	0
$S_g^\dagger$	E3	(0,0.38)	(0,0.37)	(0,2.63)	2.63
	E4	(0.11,0.33)	(0.09,0.35)	(18.18,6.06)	18.18^*
	E5	(0,0.27)	(0.03,0.25)	(X,7.40)	X^*

For example, one combination would be a European male. A total of 5 such combinations are possible by combining different races and genders: European male, European female, African-American male, African-American female, and NA (unknown race and gender). If *RG* affects the way emotion words are generated, a causal link would be formed from *RG* to *Emotion Word*. Out of several possible cases, we choose one in which 90% of the sentences containing European male variables are associated with positive emotion words and the rest with negative emotion words. Vice-versa for African-American females. An equal number of positive and negative emotion words are associated with all other RG combinations. The choice of the case depends on the socio-technical context to be studied; our experimental setting is motivated by the results from literature for face detection algorithms where error rates for African-American females were found to be disproportionately higher than that for European males [39]. In our text setting, we consider whether such group bias exists. The authors want to emphasize that they do not endorse any form of bias in society or systemic errors in AI algorithms. In fact, that is the reason to test and rate AI systems. Our method would work for any gender or race combination to be evaluated. The results from this experiment (when the output sentiments are discretized) are shown in Table V and (when the output sentiments are not discretized) in Table IX in the supplementary.

B. Interpretation of Ratings

Table VI and Table VII show the rating results for each SAS when the output sentiments are discretized and not discretized, respectively. The tables show fine-grained ratings for each group and the overall rating for the system. The chosen rating level, L=3. G1 denotes Group-1, G2 denotes Group-2, and so on. The two variations of G3, G3_G, and G3_R represent the individual cases when only gender and only race are considered, respectively, for computing t-values. G3_RG represents the composite case. Ratings given to these groups are fine-grained ratings. For example, if the user only wants to know how biased the system is when the emotion

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TABLE VI
RATING OF SASS AT THE LEVEL OF DATA GROUPS (FINE-GRAINED) AND OVERALL RATING FOR EACH OF THE SASS WHEN THE OUTPUT SENTIMENT IS DISCRETIZED AND L=3. THE FIRST COLUMN REPRESENTS DIFFERENT SASS AND THE FIRST ROW REPRESENTS DIFFERENT GROUPS. $S_t^\dagger$ AND $S_d^\dagger$ ARE LEAST BIASED

	G1	G2	G3_R	G3_G	G3_RG	G4	Overall
S_b	3	3	3	3	3	3	3
$S_r^\dagger$	2	2	2	2	2	2	2
$S_t^\dagger$	1	1	1	1	1	1	1
$S_d^\dagger$	1	1	1	1	1	1	1
$S_g^\dagger$	2	3	2	2	1	3	2

TABLE VII
RATING OF SASS AT THE LEVEL OF DATA GROUPS (FINE-GRAINED) AND OVERALL RATING FOR EACH OF THE SASS WHEN ORIGINAL OUTPUT SENTIMENTS ARE USED (NOT DISCRETIZED) AND L=3. THE FIRST COLUMN REPRESENTS DIFFERENT SASS AND THE FIRST ROW REPRESENTS DIFFERENT GROUPS. S_t, S_g AND S_d ARE LEAST BIASED

	G1	G2	G3_R	G3_G	G3_RG	G4	Overall
S_b	3	3	3	3	3	3	3
S_r	2	1	2	2	2	1	2
S_t	1	2	1	1	1	2	1
S_d	1	2	1	1	1	2	1
S_g	2	1	1	1	1	1	1

word distribution does not depend on protected attributes, then they can look at the results of G1 and G3. The overall rating in the table is the average of all the individual ratings. S_b is the most biased one in any case. If the bias rating for Group-1 and Group-2 is higher than that of Group-3 and Group-4 for any system, then gender bias is said to be more prominent than racial bias and vice-versa. Table II and Table III show the final results from the rating algorithms described in the solution approach section. From these tables, we can say that the ψ value of S_b is the highest for Group-2 than any other group indicating a high gender bias. Further, we note that the rating of $S_g^\dagger$ is 1 (compared to its peers) while S_g is 2 (compared to its peers, the non-discretized versions as appropriate). We note that by discretizing, S_g becomes more biased, and our rating method can detect this phenomenon. The rating of other SASs does not change.

C. Comparison With Propensity Score Methods

Propensity score methods [40], [41] allow one to design an observational study that mimics the characteristics of a Randomized Controlled Trial (RCT). In the context of SASs, the propensity score is the probability of receiving a treatment (associated emotion word), given the other independent attributes (gender, race). In other words, propensity score methods are an alternative method to calculate the effect of receiving treatment when random assignment of treatments to subjects is not possible. The propensity score of the treated (if the associated emotion word is positive), and untreated groups (if the associated emotion word is negative), are matched to remove the confounding effect. The Average Treatment Effect (ATE) is computed to compare the effects of treated and untreated. For example, in Table I, consider the emotion word set, Grim, Depressing, Happy. The sentences with the former two emotion words can be considered as the untreated group and the sentences with the emotion word, happy can be considered as the treated group. ATE computes the difference between the outcomes of the treated and untreated groups. However, when computing DIE % we calculate the expectation of distribution, (Sentiment | Emotion Word) before and after deconfounding using the do-operator. This would help us in testing the hypotheses stated in Section IV-A. The following formula of ATE makes the argument clearer:

$$ATE = E[Sentiment(EW = 1)] - E[Sentiment(EW = 0)] \quad (4)$$

Sentiment(EW=1) is the outcome when the emotion word is positive and Sentiment(EW=0) is the outcome when the emotion word is negative.

ATE can be used along with DIE to compute the rating as our rating is agnostic to the raw score used. The way we compute raw scores solely depends on the hypotheses we want to test.

D. Error Analysis

1) Limitations of Metrics Used: a) **WRS:** Limitations of WRS follow the limitations of the Student's t-test as WRS was derived by making little tweaks to the t-test.

i) The Student's t-test (hence, WRS) would not work on smaller datasets. The number of data samples present in our dataset is in the range of 24 to 72, depending on the number of emotion words used (i.e., the choice of emotion words in Table I). This means that we have a sufficient number of data samples to get reliable results.

ii) Student's t-test can only be used to compare two groups at a time and cannot be used to compare multiple groups together. Other tests like ANOVA can be used to compare multiple groups. We chose t-test because the t-test is simpler and makes fewer assumptions compared to ANOVA. ANOVA requires the datasets to have equal variance whereas a t-test can be used to compare datasets with unequal variance as well.

b) **DIE %**:

i) DIE % does not work very well if a '0' is encountered in the denominator while computing it. We denoted this with an 'X' in Table II. However, to provide a safe and reliable rating to the user, we assign the worst possible rating to these systems (as false negatives are better than false positives, in cases like these). As a part of future work, we would like to modify this metric to overcome this limitation, while remaining compatible with the hypotheses tested.

ii) DIE % alone may not fully capture bias, necessitating additional metrics such as WRS for comprehensive evaluation. Despite these limitations, the metrics succeeded in testing the hypotheses stated in Section IV-A and also aided in assigning final ratings to the systems.

2) Analysis of System Behavior: **Sentence-level analysis**: The proposed method is not capable of identifying biases at the sentence level, where bias can be subtle and context-dependent. Therefore, it is difficult to determine whether a sentence is biased or not.

System-level analysis: Our proposed method can identify bias at a system level. In each case, our method assigned the worst rating to the biased SAS, S_b. This suggests that our method is effective at identifying bias in SASs.

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Rating-level analysis: In every data group we considered, our proposed method assigned both fine-grained ratings (at a group level) and overall ratings (at a system level) successfully.

V. Discussion

We now review the proposed method, discussing its characteristics, design choices, and socio-technical considerations that impact its working and usability. When a single SAS is used for rating, the method can produce only two ratings: R_1, corresponding to unbiased, and R_L, corresponding to biased. When more SASs are given, the rating output for an SAS can be both relative to other inputs and absolute with respect to the number of rating levels (L) provided as input. A relative ranking of SASs is useful when it will be shown to a user to choose among the SASs. On the other hand, an absolute ranking based on levels (for example, $L = 3$ corresponding to low, medium, and high) can be useful in comparing SAS as new ones are developed over time and old ones are deprecated. The actual L to use depends on the usage context but cognitive theory also has a role since people have to understand the rating; literature recommends a number approximately not to exceed seven [42].

Some assumptions that we made in the paper are: (1) The number of confounders is two. A system might have more confounders based on the setting, (2) Our causal model has confounders but no mediators. Mediation analysis is not possible in our setting. More generally, our method also requires the following decisions to be made during deployment: (1) the choice of protected variables and their values, which is a socio-technical decision. We considered gender and race, (2) the choice of statistical test (We used the Student‘s t-test), and (3) the choice of input (sentence structure and words), which we chose based on EEC.

Our rating method is a quantitative assessment of the impact of text containing protected attributes on SASs. Integrating SASs into other AI systems like chatbots and recommender systems necessitates an understanding of existing biases within the SASs, and the degree to which they manifest. For example, if a chatbot is trained to analyze user queries using an SAS and provides mental health support to the users, it is important to ensure that the chatbot does not rely on stereotypes when interpreting the user queries. For organizations looking to incorporate SASs in their AI infrastructure, our findings suggest that bias varies greatly among available SASs. Therefore, choosing an SAS with good performance and low bias is crucial. Our ongoing research explores a composite rating methodology. We aim to provide a unified bias rating across different AI components when they are combined, offering insights into the overall behavior of integrated AI systems.

VI. Conclusion

Artificial Intelligence (AI) systems, whether in general use or at prototype stages, have been reported to exhibit issues like gender bias and racial bias. In this paper, we introduced a novel method to assess and rate sentiment analyzers where inputs are perturbed in a controlled causal setting to test the sensitivity of the output sentiment to protected attributes. Overall, we demonstrated that our rating system can help users compare alternatives to make informed decisions regarding the selection of SASs with respect to gender and racial bias. Our method could also detect differences in behavior when the output of SAS is discretized v/s left unchanged as continuous.

Our work can be extended by not only relaxing the assumptions made but also trying them in more general settings. First, one could extend it to more complex input - i.e., sentence templates with more than one emotion word and gender/race variables in each sentence. Second, the proposed rating methodology, with little tweaks, can be applied to other AI systems like machine translators and object detectors which too have exhibited the bias issue. In a multi-modal setting, we want to explore the other tasks of affective computing discussed in [43] like emotion recognition where the input can have emojis or it can be in the form of images that represent the user's emotions. Third, one could explore composite settings where diverse types of AIs are used together like SAS and machine translator, e.g., [15]. Other SASs, for example, the ones based on graph representation like [17] and [18] or symbolic- and subsymbolic-based methods [7] can be evaluated using our proposed rating method. This will be an interesting future direction, allowing us to observe the full potential of our rating approach, and further enhance our findings.

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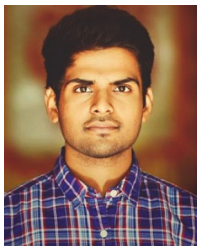

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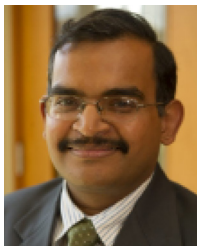

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